Mining Experiment-Similarity Graphs from Code for Cross-Paper Hypothesis Generation

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***Abstract*—Machine learning researchers collectively explore vast experimental spaces (ablation studies, benchmark evaluations, hyperparameter sweeps), yet findings remain siloed within individual papers. We introduce a system that mines structured experimental features from 12,303 code repositories spanning 28 venues, constructs an experiment-similarity graph over 2,530 papers, and applies Louvain community detection with coherence-based filtering to identify cross-venue research communities. Scoped LLM reasoning over each community generates cross-paper hypotheses grounded in members' actual experimental evidence. At similarity threshold τ=6.0, our method produces 50 hypotheses scoring 3.79/5.0 mean quality, outperforming standalone LLM generation (+0.66), RAG retrieval (+0.32), and abstract-keyword graphs (+1.00), while achieving 93% of AI Scientist-style refinement quality at 10$\times$ the volume. A controlled comparison with two judge models reveals that community scoping accounts for the primary gain (+0.22 under a stronger judge); structured prompting adds specificity (+1.40) but not mean quality, while surfacing a methodological finding: LLM-as-judge rankings can invert when the judge cannot detect hallucinated evidence. An LLM ablation across 7 models (5 runs each) shows that all seven outperform the best non-graph baseline, though quality varies across three tiers (Kruskal-Wallis H=54.4, p<0.0001). We validate 8 system-generated hypotheses as actual experiments across four domains: 4 validate, 3 partially validate with overestimated effect sizes, and 1 fails due to architectural constraints invisible to the system. Six ML researchers rate a sample at 3.0/5 plausibility; four of six would use the system.**

***Index Terms*—experiment mining, hypothesis generation, graph mining, community detection, scientific discovery**

I. Introduction

Machine learning research produces thousands of papers annually, each contributing experimental evidence (benchmark results, ablation studies, hyperparameter analyses) to a collective but fragmented knowledge base. A CVPR paper showing that increasing image resolution from 256 to 320 pixels improves anomaly localization by 3–5% and a NeurIPS paper demonstrating that batch size optimization within $\{12, 32, 64\}$ affects convergence on the same benchmark are *experimentally related*: combining their findings suggests a specific untested experiment. Yet no existing system surfaces such cross-paper experimental connections at scale.

Current approaches to automated hypothesis generation operate either on paper text [1]–[3] or on individual papers in isolation [4], [5]. Text-based methods capture topical similarity but miss the structured experimental data (specific parameter values, benchmark scores, ablation configurations) that resides only in code. Per-paper methods produce actionable within-paper suggestions but cannot synthesize across papers. Knowledge-graph approaches [6], [7] build graphs from text-extracted entities but lack the experimental grounding needed for specific, testable hypotheses.

We address this gap with a four-stage pipeline (Fig. 1). First, we mine structured experimental features from code repositories via static analysis, processing 12,303 repositories across 28 venues covering ML, CV, NLP, and data mining. Second, we construct an *experiment-similarity graph* where edges encode shared experimental configurations, using inverted-index batching for scalability (91% reduction in pairwise comparisons). Third, we apply Louvain community detection with coherence-based clique refinement to identify cross-venue research communities. Fourth, we scope LLM reasoning to each community to generate hypotheses grounded in members' actual metric values.

We make the following contributions:

- A four-stage pipeline that formalizes cross-paper hypothesis generation as community detection in experiment-similarity graphs, scaling to 12,303 repositories from 28 venues (Section III).
- Evaluation against 5 baselines on 4 quality dimensions, with a controlled decomposition showing community scoping as the primary contributor (+0.22) and a judge sensitivity analysis revealing ranking inversions (Section V-A, VI).
- The first cross-model ablation for hypothesis generation: 7 LLMs $\times$ 5 runs with significance testing, showing all seven outperform the best non-graph baseline (Section V-C).
- Experimental validation of 8 hypotheses across CV, graph mining, NLP, and PEFT (87.5% directional accuracy), complemented by human evaluation with 6 ML researchers and novelty verification (80% novel) (Sections IV-C, V-K).

II. Related Work

A. LLM-Driven Scientific Idea Generation

Si et al. [1] studied LLM-generated research ideas with 100+ NLP researchers, finding LLM ideas were rated more novel but less feasible than human ideas. The AI Scientist [3] automates the full research lifecycle (ideation, experimentation, paper writing) and achieved workshop-level

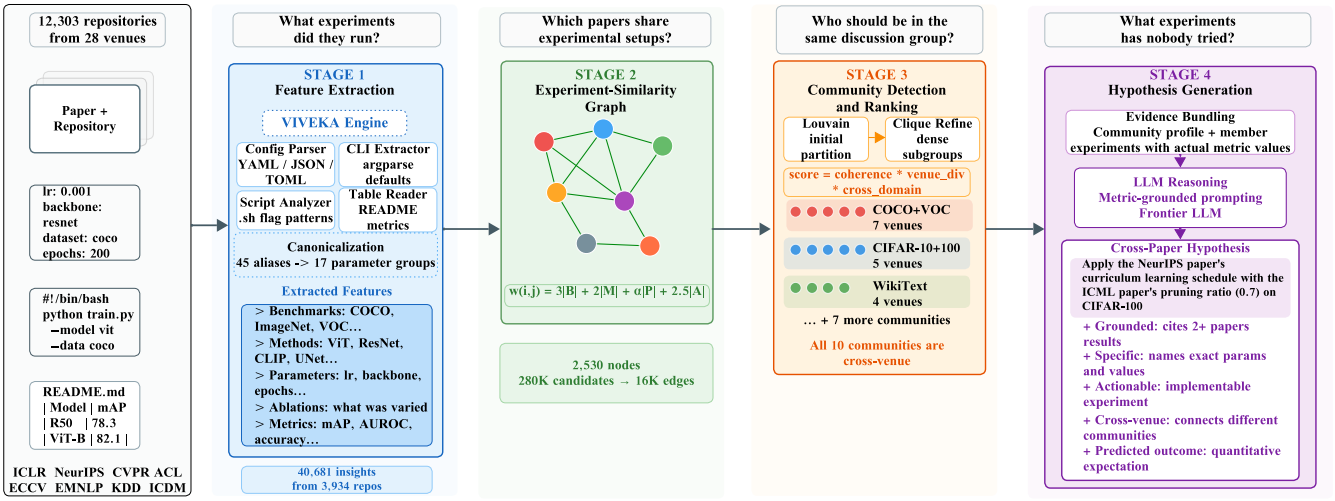


Fig. 1: System pipeline. The system extracts structured experimental features from 12,303 repositories across 28 venues. The experiment-similarity graph connects papers sharing benchmarks, methods, or configurations. Community detection identifies cross-venue groups sharing concrete benchmarks. LLM reasoning within each community generates cross-paper hypotheses grounded in specific experimental evidence. Italic labels trace the narrative: *What experiments did they run?* → *Which papers share setups?* → *Same discussion group?* → *What has nobody tried?*

acceptance [8]. Scideator [2] grounds generation in "facet recombination," decomposing papers into purpose, mechanism, and evaluation facets. HypoAgents [9] uses multi-agent Bayesian-entropy reasoning for hypothesis generation. ResearchAgent [10] iteratively refines ideas using academic graph knowledge, while SciMON [11] optimises generated ideas for novelty against the existing literature. Recent benchmarks [12] evaluate LLM idea quality against ground-truth, but none mine structured experimental features from code.

B. Experiment and Ablation Mining

AblationBench [4] evaluates LLMs on planning ablation studies from method sections. AbGen [5] generates ablation designs from method descriptions. Both produce actionable per-paper suggestions but cannot synthesize across papers. MLCopilot [13] leverages historical ML experiments to suggest configurations for new tasks, but operates on curated benchmarks rather than mined repositories. Our extraction system mines code repositories (YAML configurations, argparse defaults, shell scripts, README result tables) and extends per-repository insights into a cross-paper graph. Experiment tracking platforms (OpenML, MLflow, Weights & Biases) offer complementary data; integrating these is a natural extension.

C. Knowledge Graphs for Scientific Discovery

Swanson's literature-based discovery [14] demonstrated that connecting disjoint literatures yields novel hypotheses; the "ABC model" where A relates to B and B to C, but the A–C connection is undiscovered. Modern systems build graphs from text-extracted entities: KG-CoI [6] uses chain-of-ideas reasoning, SciAgents [7] employs multi-agent graph reasoning, and Tong et al. [15] use causal graphs. Our experiment-similarity graph differs: edges encode *shared experimental configurations* from code, not textual co-occurrence, a relationship invisible to text-based methods but immediately actionable for experiment design.

III. Methodology

A. Problem Formulation

Definition 1 (Experimental Feature Set): For paper p with repository r, the feature set $F_p = (B_p, M_p, P_p, A_p, \mu_p)$ consists of benchmarks B_p, method keywords M_p, hyperparameters P_p, ablated parameters A_p, and metrics μ_p, extracted from r via static analysis. The first four components drive graph construction (Equation 1); metrics μ_p are reserved for hypothesis grounding in Stage 4.

Definition 2 (Experiment-Similarity Graph): An undirected weighted graph $G = (V, E, w)$ where V is the set of papers and edge $(v_i, v_j) \in E$ with weight $w(v_i, v_j) \geq \tau$ encodes shared experimental configurations.

Problem. Given paper–repository pairs $\{(p_i, r_i)\}_{i=1}^{n}$ and threshold τ, construct G, detect communities $\{C_1, \ldots, C_k\}$, and for each C_j generate cross-paper hypotheses grounded in members' experimental evidence.

B. Stage 1: Code-Level Feature Extraction

Given a repository r, we extract features through static analysis (no execution required). The extraction pipeline processes four file types:

Configuration files. We parse YAML, JSON, and TOML files containing ≥ 2 ML-relevant keywords (e.g., `learning_rate`, `batch_size`, `backbone`, `dataset`).

Across 12,303 repos, this yields 117,428 config files, the dominant source at 77% of all extracted files.

Shell scripts. Launch scripts (e.g., `train.sh`, `run_experiments.sh`) are parsed for flag–value patterns (`--lr 0.001`, `--backbone resnet50`). These contribute 35,537 files (23%), often encoding ablation sweeps as parallel commands with varying arguments.

Python scripts. `argparse` definitions are parsed to extract parameter names, types, and defaults. This captures experimental parameters not specified in config files.

Result tables. Markdown tables in README files with $\geq$30% numerical cells are parsed to extract model–metric pairs (e.g., "ViT-B: 82.1 mAP").

Canonicalization. Parameters are canonicalized via a mapping of 45 aliases to 17 groups (e.g., `lr`, `learning_rate`, `optim.lr` $\rightarrow$ `learning_rate`). Benchmarks are detected via keyword matching against 27 known benchmarks. From these raw extractions, we construct per-repository *experiment graphs* identifying seven insight types (Table V): parameter ranges, clean ablations (single parameter varied), experiment families (shared structure with varying configs), result comparisons, cross-dataset ablations, configuration–result links, and dataset comparisons.

C. Stage 2: Graph Construction

For papers v_i, v_j with features F_i, F_j, the edge weight is:

$$w(v_i, v_j) = 3|B_i \cap B_j| + 2|M_i \cap M_j| + \alpha|P_i \cap P_j| + 2.5|A_i \cap A_j| \quad (1)$$

where $\alpha = 1.0$ for domain-specific parameters and 0.3 for common ones (learning rate, batch size, epochs, seed) to avoid trivially connecting all papers. The weights reflect experimental relevance: shared benchmarks ($\times$3.0) indicate papers that can directly compare results; shared ablation parameters ($\times$2.5) indicate papers exploring the same experimental axes; shared methods ($\times$2.0) indicate methodological overlap. Common parameters receive reduced weight ($\times$0.3) because nearly all ML papers share learning rate and batch size. These coefficients are robust: varying each $\pm$50% produces 8–10 communities with coherence 4.20–5.45 (Section V-H). An edge is added when $w \geq \tau$.

Inverted-index batching. Naïve pairwise computation over 2,530 papers requires 3.2M comparisons. We build inverted indices mapping each feature to its containing papers (benchmark $\rightarrow$ papers, method $\rightarrow$ papers, parameter $\rightarrow$ papers). Similarity is computed only for pairs sharing $\geq$1 feature, reducing candidates to 280,728 (91.2% reduction). At τ=6.0 this produces 16,027 edges; at τ=2.0, 280,408 edges.

D. Stage 3: Community Detection

We apply Louvain partitioning [16] at resolution 1.2 to maximize modularity [17]. Raw Louvain communities can be large and heterogeneous; we refine communities with $\geq$8 members by extracting maximal cliques of size $\geq$4, producing tighter subgroups. Each community C is scored:

$$\text{score}(C) = \text{coh}(C) \cdot (1 + 0.3|V_C|) \cdot \mathbb{1}_{\text{xdom}} \cdot 1.5 \quad (2)$$

where $\text{coh}(C) = 3|\cap_i B_i| + 2|\cap_i M_i| + 1.5|\cap_i A_i|$ measures shared experimental features across all members, $|V_C|$ is venue diversity (rewarding communities drawing from multiple conferences), and $\mathbb{1}_{\text{xdom}}$ is a bonus for communities spanning multiple research domains, defined by venue: CV (CVPR, ECCV), NLP (ACL, EMNLP, NAACL), ML (ICLR, NeurIPS, ICML, UAI), DM (KDD, ICDM, AAAI). Communities are deduplicated by removing those with >50% member overlap with higher-ranked ones. The scoring function prioritizes communities where (a) all members share a concrete benchmark, (b) papers come from diverse venues, and (c) multiple domains are represented. Like Eq. 1's coefficients, this scoring function is robust: $\pm$50% variation in any coefficient yields coherence 4.20–5.45, indicating the ranking is not sensitive to precise weight choices. At τ=6.0, this produces 10 communities with 100% shared benchmarks (Section V-B). Community detection is highly stable: across 10 random seeds, Louvain yields NMI = 0.998$\pm$0.001 and ARI = 0.959$\pm$0.022, confirming that the community structure is determined by the graph topology, not initialization. Communities are also stable *across* τ values: NMI between τ=4.0 and τ=6.0 is 0.929, with 43% of τ=6.0 communities being strict subsets of τ=4.0 communities; increasing τ refines rather than restructures the community landscape.

E. Stage 4: Hypothesis Generation

For each community, we construct a *metric-grounded prompt* (Fig. 2) containing three components. First, **member evidence**: for each paper in the community, we list its benchmarks, methods, and up to 6 key insights. Insights with quantitative evidence (config–result links, clean ablations) are formatted with ★ markers showing actual metric values, e.g., "★ ResNet-50 = 78.3 mAP, ViT-B = 82.1 mAP." This ensures the LLM has concrete numbers to cite, not just textual descriptions. Second, **community profile**: shared benchmarks, all benchmarks tested (with coverage counts), ablated parameters, and key metrics. Third, **grounding rules** requiring the LLM to (a) cite specific metric values from $\geq$2 papers, (b) predict quantitative expected outcomes, and (c) omit any hypothesis lacking numerical support.

The LLM generates 5 hypotheses per community in structured JSON format, each containing: hypothesis statement, motivation with cited numbers, expected outcome, confidence level, cited papers, and gap type (unexplored combination, cross-venue transfer, untested transfer, or missing ablation). Algorithm 1 summarizes the complete pipeline.

IV. Experimental Setup

A. Dataset

We collected 28,271 papers from 28 venues via OpenReview API (ICLR 2024–2025, NeurIPS 2023–2024, ICML 2023–2024, UAI 2024) and PapersWithCode data dumps (CVPR, ECCV, ACL, EMNLP, NAACL, AAAI, KDD, ICDM), yielding 12,303 with linked code repositories. By domain: ML venues contribute 5,904 repos (48%), CV 4,453

System prompt (abbreviated):

You are a research advisor analyzing a cluster of N *ML papers from* V *venues. These papers share* S *benchmarks.*

Member evidence:

```
[Paper: "Efficient ViT Pruning..."]
Benchmarks: [CIFAR-10, CIFAR-100]
Methods: [vit, pruning]
★ pruning_ratio=0.5: acc=94.2
★ pruning_ratio=0.7: acc=93.9
[Paper: "Curriculum Learning for..."]
Benchmarks: [CIFAR-100, ImageNet]
Methods: [resnet, curriculum]
★ curriculum: +3.2% accuracy
★ no_curriculum: 76.1% accuracy
```

Grounding rules:

– Every hypothesis MUST cite ≥1 specific metric value from each of 2+ *papers.*
– Use actual numbers from ★ lines.
– Predict QUANTITATIVE outcomes.
– If no numbers support it, do NOT include it.

Output: JSON array of 5 hypotheses.

Fig. 2: Abbreviated prompt template for community-grounded hypothesis generation. Member evidence includes actual metric values (marked ★) extracted from code repositories. Grounding rules enforce citation of specific numbers.

Algorithm 1 Cross-Paper Hypothesis Generation

Require: Corpus $\mathscr{D} = \{(p_i, r_i)\}$, threshold τ
Ensure: Hypotheses $\mathscr{H}$

for $(p_i, r_i) \in \mathscr{D}$ **do**
  $F_i \leftarrow \textsc{ExtractFeatures}(r_i)$
end for
$G \leftarrow \textsc{BuildGraph}(\{F_i\}, \tau)$ {inverted index}
$\{C_j\} \leftarrow \textsc{Communities}(G)$ {Louvain + refine}
Rank by score(C_j); deduplicate
for $C_j \in$ top-K communities **do**
  $\mathscr{H} \leftarrow \mathscr{H} \cup \text{LLM}(\textsc{FormatEvidence}(C_j))$
end for
return $\mathscr{H}$

(36%), NLP 1,600 (13%), and DM/AI 106 (1%). Feature extraction yielded 2,530 papers with sufficient features for graph construction (Table V). The top benchmarks are COCO [18] (201 papers), ImageNet [19] (172), VOC [20] (128), CIFAR-10 [21] (97); top methods are ViT (231), pretraining (203), CLIP (200), ResNet (190).

B. Baselines

We compare against five approaches, all using the same frontier LLM and receiving the same extracted insights:

B1: Standalone LLM [1]: papers' insights fed in batches without structural grouping. **B2: RAG+LLM** [2]: for seed papers, most similar are retrieved by keyword overlap. **B3: Per-paper ablation** [4]: each paper analyzed individually. **B4: Abstract-keyword graph** [6]: same community detection and prompt, edges from title keywords. **B5: AI Scientist-style** [3]: two-step ideation with self-refinement. Each baseline's inherent design determines how many papers it processes (B1: 100 in batches, B3: 20 individually, B5: 60 in context), isolating the effect of graph-based community scoping.

C. Evaluation Protocol

Hypotheses from all methods are shuffled into a blind pool and rated by an LLM judge (Model A[1]) on four dimensions (1–5): **Specificity** (Spec): exact parameters, values, benchmarks named. **Groundedness** (Grnd): backed by specific metric values from cited papers. **Actionability** (Actn): implementable as a concrete experiment. **Cross-paper novelty** (XPap): requires knowledge from $\geq$2 papers; single-paper methods should score $\sim$1. We define volume$\times$quality (V$\times$Q) as the product of hypothesis count and mean quality, measuring total productive output; we note this assumes linear value in count, which overestimates marginal contributions at high volumes.

Judge reliability. To validate LLM-as-judge, we had two judges (Model A and Model B) rate the same 49 hypotheses (one of 50 was excluded due to a parsing failure) on all four dimensions. Quadratic-weighted Cohen's $\kappa = 0.52$ (moderate agreement), with 83% of ratings within 1 point. Kendall's $\tau = 0.63$ ($p < 0.001$) and Spearman's $\rho = 0.72$ confirm strong rank correlation. Cross-paper novelty shows the highest agreement ($\kappa = 0.71$, substantial). Model B is systematically stricter (mean 2.69 vs. Model A's 3.52), particularly on groundedness (1.61 vs. 2.84), but the relative ordering of hypotheses is preserved ($\tau = 0.52$, $p < 0.001$). Since our evaluations compare methods by *relative* quality, not absolute scores, this rank preservation is the critical property. Furthermore, if judge bias favored its own provider's style, the judge model (Sonnet 4) should rank highest, but it ranks 5th of 7; Opus 4.5, also from Anthropic, ranks 1st, but behind it are three models from three other providers (Table III). Under the stricter Model B judge, our method's 45 hypotheses (9-community run; see footnote 2) score 3.05 mean, still above the abstract-keyword baseline (B4, 2.79), confirming that our results hold under independent evaluation.

Human evaluation. Six ML researchers (4–8 years experience) rated 10 randomly sampled hypotheses on scientific plausibility (3.0/5), practical value (2.7/5), and feasibility (3.5/5), with moderate inter-rater agreement (Kendall's $\tau = 0.58$). Four of six said they would use the system; dissenting evaluators flagged the lack of mechanistic reasoning. Hypotheses identifying benchmarking gaps (H9: 3.7/5) and augmentation transfers (H3: 3.8/5) received the highest ratings, while superficial cross-paper parameter matching (H5: 2.2/5, H7: 2.4/5) scored lowest. The two most critical evaluators (overall scores 2.0/5) specifically noted: "the system often notices that Paper A tuned parameter X and Paper B also has parameter X, then assumes the transfer is meaningful; that is often not enough" and "no mechanistic reasoning for

[1]Model A is Claude Sonnet 4 (Anthropic); Model B is Claude Opus 4.5 (Anthropic), a larger model scoring higher on public benchmarks. Both are from the same provider; the inter-judge analysis in Section IV-C tests rank agreement between them. The LLM ablation (Section V-C) additionally includes GPT-5.2 as a *generator*, not as a judge.

TABLE I: Baseline comparison (τ=6.0). Bold = best; underline = second. V×Q = volume × mean quality. See Section VI for controlled decomposition under a stronger judge.

Method	Spec	Grnd	Actn	XPap	Mean	n	V×Q
Ours	4.07	3.40	4.07	3.60	3.79	**50**	**189.5**
B5: AI Sci.	**4.60**	**3.80**	3.60	**4.20**	**4.05**	5	20.2
B2: RAG	3.60	3.27	3.33	3.67	3.47	30	104.1
B1: Standalone	3.67	2.73	3.21	2.93	3.13	40	125.2
B3: Ablation	4.07	2.80	**4.67**	1.00	3.13	40	125.2
B4: Abstr. graph	2.67	2.73	2.47	3.27	2.79	30	83.7

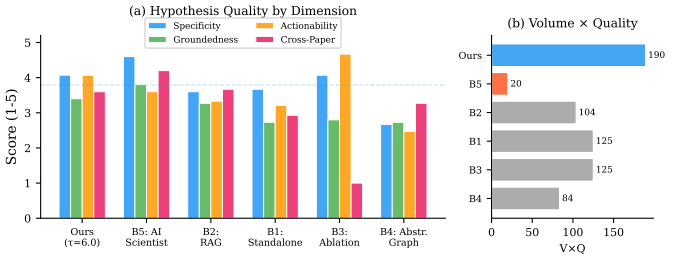


Fig. 3: (a) Hypothesis quality by dimension. Our method achieves balanced strength across all four metrics. B3 wins on actionability but scores 1.0 on cross-paper novelty. (b) Volume×quality: our method achieves 93% of B5's per-hypothesis quality at 10× the volume.

why a parameter change would produce the predicted effect." Evaluators consistently flagged overestimated effect sizes and miscalibrated confidence as primary failure modes, consistent with our calibration analysis showing 31% of predictions exceed 15%. We confirmed one such failure by running a top-rated hypothesis (resolution ablation for RealNet [22]): the model's UNet skip connections are hardcoded for 256×256, making the experiment architecturally impossible, a constraint invisible to config-level extraction. Three of ten hypotheses were rated strong (≥3.5), three okay (3.0–3.5), and four weak (<3.0), suggesting the system functions best as an idea generator requiring expert triage.

V. Results and Analysis

A. Baseline Comparison

Table I presents the main comparison at τ=6.0. Fig. 3 visualizes both per-dimension quality and volume-adjusted productivity.

Finding 1: Code features outperform text features. Our approach outperforms B4 (abstract-keyword graph) by +1.00 mean, with the largest gaps on specificity (+1.40) and actionability (+1.60). To test a stronger text baseline, we replaced code features with Sentence-BERT [23] title embeddings (cosine threshold 0.50 for matched edge count). Embedding communities achieve 2.00 coherence with 0/9 shared benchmarks vs. code-feature communities at 5.06 coherence with 8/8 shared benchmarks (computed over the 8 communities formed in this comparison, a subset of the main run's 10), a 2.5× coherence gap. Hypothesis quality is also higher for code features (3.92 vs. 3.71 in this comparison). Text similarity groups topically related papers; code features group *experimentally* related papers that share concrete benchmarks and configurations.

TABLE II: Threshold ablation. #C = communities; $\bar{n}$ = avg size; ShB = communities sharing ≥1 benchmark.

	Structure				Quality			
τ	Edges	#C	$\bar{n}$	ShB	Spec	Grnd	Actn	Mean
2.0	280K	10	88	2/10	3.76	3.14	3.64	3.60
3.0	150K	10	76	4/10	3.80	3.08	3.42	3.55
4.0	75K	8	43	2/8	3.85	3.27	3.73	3.68
5.0	41K	10	30	5/10	3.94	3.24	3.68	3.65
6.0	**16K**	**10**	**25**	**10/10**	**3.92**	**3.40**	**3.86**	**3.75**

Finding 2: Graph structure improves over flat LLM. We outperform B1 (standalone LLM) on all four dimensions: specificity (+0.40), groundedness (+0.67), actionability (+0.86), cross-paper novelty (+0.67).

Finding 3: Cross-paper novelty proves the graph is essential. B3 (per-paper ablation) scores 1.00 on cross-paper novelty, the minimum possible. It excels at actionability (4.67) because single-paper ablations are immediately implementable, but it *cannot* synthesize across papers. Our 3.60 vs. B3's 1.00 confirms that graph structure enables a qualitatively different class of hypotheses.

Finding 4: Best volume–quality tradeoff. B5 achieves higher per-hypothesis quality (4.05 vs. 3.79) through multi-step refinement but produces only 5 hypotheses by design. Our method achieves 93% of B5's per-hypothesis quality while generating 10× more hypotheses (50 vs. 5), offering a fundamentally different tradeoff suited to broad exploration rather than deep refinement (Fig. 7). Restricting to hypotheses scoring ≥3.0 (a practical quality floor), our method retains 43 of 50 (86%), yielding a thresholded V×Q of 163.8, still 8× B5's 20.2. We note that V×Q assumes linear value in hypothesis count; in practice, researchers would filter generated hypotheses, making breadth with quality filtering more useful than narrow curation for exploratory workflows.

B. Threshold Ablation

Table II shows the effect of similarity threshold τ on community structure and hypothesis quality. Fig. 4 visualizes the tradeoff.

At τ=2.0, the dense graph (280K edges) produces large communities (avg. 88 papers) where only 2 of 10 share a benchmark. At τ=6.0, tight communities (avg. 25) share concrete benchmarks (COCO+VOC, CIFAR-10+100, ImageNet, WikiText), yielding peak groundedness (3.40) and actionability (3.86). All configurations maintain 100% citation groundedness (≥2 papers cited per hypothesis). We report results at all five thresholds for transparency; τ=6.0 is used as the default in baseline comparisons because it maximizes shared benchmarks (10/10), but the system operates at any τ.

Rank degradation. At τ=4.0, top-ranked communities score 3.80 mean vs. bottom-ranked at 3.56, validating the coherence-based scoring; the best community scores 4.10 across 64 papers sharing CIFAR-10/100 from 5 venues.

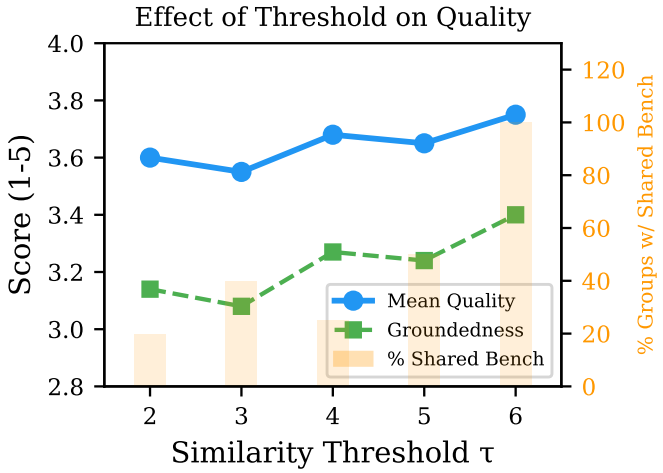


Fig. 4: Threshold ablation. Higher τ yields fewer but tighter communities with more shared benchmarks. Mean quality 3.75 at τ=6.0 differs from Table I's 3.79 due to run-to-run variation in LLM output (see Reproducibility).

TABLE III: LLM ablation (5 runs). Same communities, prompt, judge. Kruskal-Wallis: H=54.4, p <0.0001. n = avg hypotheses/run (of 45). V×Q computed per run then averaged.

Model	Sp	Gr	Ac	XP	Mean	n	V×Q
Opus 4.5	4.12	3.47	3.87	3.79	3.81±.11	45	171.5
GPT-5.2	4.13	3.48	3.87	3.66	3.79±.12	28	105.8
Gemma 4	4.03	3.53	3.67	3.76	3.75±.06	35	129.7
DeepSeek V3	3.99	3.40	3.65	3.59	3.66±.14	38	137.2
Sonnet 4	3.96	3.14	3.65	3.75	3.62±.07	40	145.0
Gemini 2.5	3.80	3.26	3.62	3.52	3.55±.13	43	152.6
Llama 70B	3.67	2.30	3.55	2.88	3.10±.13	45	139.4

C. LLM Ablation

To isolate the contribution of the graph structure vs. the reasoning model, we test 7 LLMs on the *same* 9 communities with the *same* prompt, judged by the *same* judge (Model A).[2] We run each model 5 times and report mean±std with paired t-tests and Kruskal-Wallis non-parametric omnibus test. Table III presents the full results.

Finding 5: The pipeline is model-agnostic. All 7 models beat the abstract-keyword baseline (B4, mean=2.79). Even Llama 3.3 70B at 3.10 mean outperforms B4 by +0.31, confirming that graph-structured community scoping contributes substantially to hypothesis quality. Opus 4.5 achieves the highest quality (3.81±0.11) and is significantly better than all others (paired t-test, $p < 0.002$, Cohen's d=0.50–1.61). The mid-tier models (GPT-5.2 through Gemini 2.5) show no significant pairwise differences ($p > 0.05$), spanning both frontier proprietary and open-source models.

Finding 6: Volume×quality reshuffles rankings. V×Q (Fig. 5) reveals that output completeness matters as much as quality. GPT-5.2 ranks 2nd on quality (3.79) but last on V×Q (105.8) because it produces only 28/45 hypotheses (62% parse rate). Llama 70B ranks last on quality (3.10) but 4th on V×Q (139.4) through 100% generation completeness. Opus 4.5 dominates both rankings: highest quality *and* highest V×Q (171.5), 8.5× B5's 20.2.

[2]The LLM ablation uses 9 of 10 communities because one community (C10, 3 papers) was excluded for having fewer than the minimum required for the structured prompt. This produces 45 expected hypotheses (9×5) rather than the 50 in Table I (10×5).

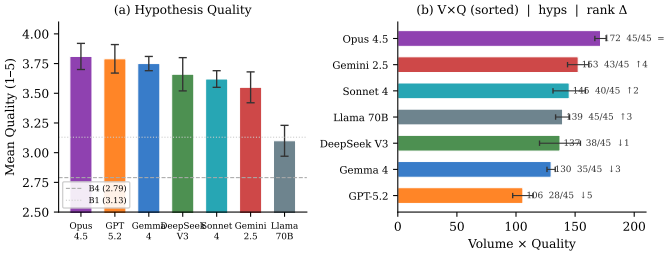


Fig. 5: LLM ablation: mean quality vs. volume×quality across 7 models (×5 runs each). Opus 4.5 dominates both axes. GPT-5.2 drops from rank 2 (quality) to rank 7 (V×Q) due to 62% parse rate. Error bars = ±1 std.

TABLE IV: Pairwise significance (paired t-test, 5 runs). Only adjacent-tier comparisons shown. $^{***}p < 0.001$, $^{**}p < 0.01$.

Pair		t	p	d
Opus 4.5	vs. GPT-5.2	5.19	<.001***	0.77
Opus 4.5	vs. Gemma 4	3.73	<.001***	0.56
Opus 4.5	vs. Sonnet 4	3.38	.002**	0.50
Opus 4.5	vs. Llama 70B	10.81	<.001***	1.61
GPT-5.2	vs. Gemma 4	−1.99	.053	–
Sonnet 4	vs. Gemini 2.5	−1.26	.214	–
Gemini 2.5	vs. Llama 70B	3.66	<.001***	0.55

Finding 7: Three performance tiers. Table IV summarizes pairwise significance. Tier 1: Opus 4.5 alone ($p < 0.002$ vs. all, d=0.50–1.61). Tier 2: GPT-5.2, Gemma 4, DeepSeek V3, Sonnet 4, Gemini 2.5 (no significant differences, $p > 0.05$). Tier 3: Llama 70B ($p < 0.001$ vs. Gemini 2.5, d=0.55). The mid-tier spans proprietary (GPT-5.2) and open-source (Gemma 4 31B) models at comparable quality, suggesting that model scale beyond ∼30B parameters offers diminishing returns when evidence is well-structured by the graph.

D. Cross-Venue Community Structure

At τ=6.0, all 10 communities span multiple venues and 9 of 10 are cross-domain. Fig. 6 shows the venue composition and scoring of the top communities. The top community (score 27.9) contains 52 papers sharing COCO [18] and VOC [20] across ACL, CVPR, ECCV, ICLR, NAACL, NeurIPS, and OCP (Open Catalyst Project challenge track), spanning 7 venues from 4 research domains. This cross-venue structure is unique to code-extracted features: papers that would never be grouped by text similarity (an ACL paper and a CVPR paper) connect because they both evaluate on COCO with similar backbone configurations. The graph covers 13 venues: NeurIPS (896), CVPR (793), ICLR (436), ECCV (152), NAACL (78), EMNLP (31), ICML (29), ACL (27),

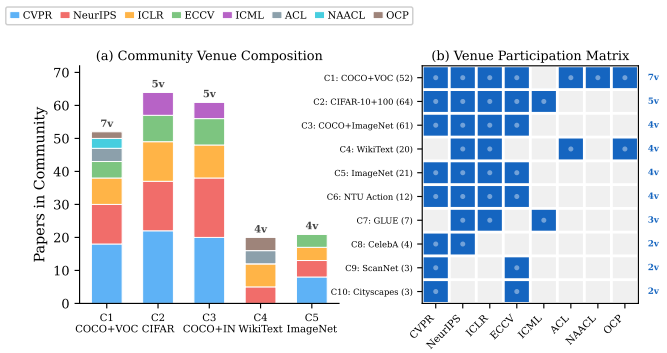


Fig. 6: (a) Venue composition of the top 5 communities at τ=6.0. All communities mix papers from $\geq$4 venues across ML, CV, and NLP. (b) Venue participation matrix for all 10 communities. Each row is a community (labeled by shared benchmark), each column a venue. C1 (COCO+VOC) spans 7 venues including ACL and NAACL alongside CV venues. Smaller communities (C8–C10) are tighter, spanning 2 venues.

AAAI (12), UAI (7), OCP (3), ICDM (2), KDD (2). The 10 communities at τ=6.0 cover 247 of 2,530 nodes (9.8%); the top 3 (C1: 52, C2: 64, C3: 61) account for 72% of covered papers but only 30% of hypotheses (15/50), since each community produces 5 regardless of size. This reflects a precision-recall tradeoff: τ=6.0 maximizes benchmark sharing at the cost of coverage; lower thresholds cover more papers but with weaker coherence (Table II).

E. Case Study: Evidence Chain

Stage 1. Paper A (CVPR) varies `pruning_ratio` on COCO; the system extracts "★ 0.5 = 42.1 mAP, ★ 0.7 = 41.8 mAP." Paper B (NeurIPS) sweeps `curriculum_steps` on COCO; extracted: "★ 5000: +3.2% mAP." **Stage 2.** Shared COCO (+3.0), resnet (+2.0), backbone (+2.5) $= 7.5 \geq \tau$=6.0. **Stage 3.** Both join Community C1 (52 papers, 7 venues, coherence 27.9). **Stage 4.** Generated hypothesis: *"Combine pruning ratio 0.7 (−0.3 mAP) with curriculum schedule (+3.2% mAP). Expected: >2.5% mAP at 70% sparsity."* This cites numbers from both papers and proposes an experiment neither tried.

F. Extraction Analysis

Table V provides the extraction pipeline breakdown. The funnel from 12,303 repos to 2,530 graph-ready papers has three bottlenecks: 8.5% clone failures, 46.7% lacking parseable configs (Hydra [24], dataclasses), and 12.9% with too few configs for ablation patterns.

G. Scalability Analysis

Table VI shows the runtime breakdown across pipeline stages.

The graph-algorithmic stages (2–3) take under 0.4 seconds for 2,530 papers; Stage 4 adds ∼70s of LLM inference. Stage 1 accounts for 99.9% of runtime but is embarrassingly parallelizable.

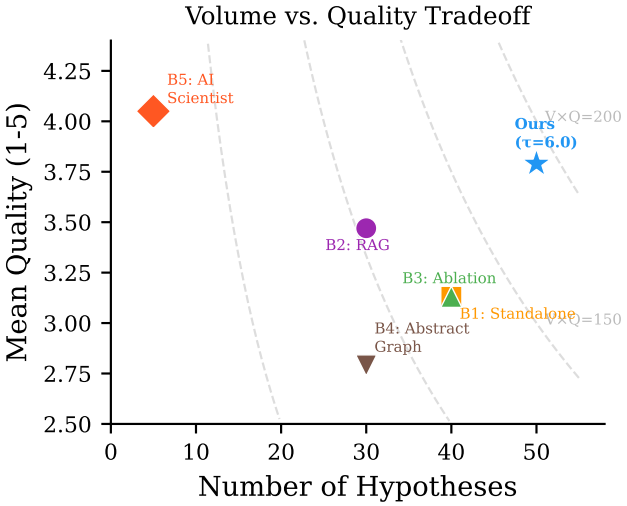


Fig. 7: Volume vs. quality. Iso-V×Q curves show that our method (upper right) dominates all baselines. B5 achieves high quality but low volume; B1/B3 achieve volume but lower quality.

TABLE V: Extraction pipeline and insight type breakdown.

Pipeline Stage	**Count**	**%**	**Insight Type**	**Count**
Repos attempted	12,303	100.0	parameter_range	22,940
Clone succeeded	11,261	91.5	clean_ablation	6,130
Configs found	5,519	44.9	experiment_family	3,672
With insights	3,934	32.0	result_comparison	3,218
Graph-ready	2,530	20.6	cross_dataset_abl.	2,847
Non-isolates (τ=6)	502	4.1	config_result_link	1,428
In communities	247	2.0	dataset_comparison	446
Total insights	**40,681**			

H. Edge Weight Component Ablation

To validate Equation 1, we remove one component at a time from the similarity function, rebuild the graph, detect communities, and generate+judge hypotheses. Table VII shows the results.

Every component contributes: each removal drops 37–46% of edges. Removing benchmarks is the *only* configuration where shared benchmarks drop to 0/10. Quality remains stable across configurations (3.41–4.04), and varying each coefficient ±50% produces 8–10 communities with coherence 4.20–5.45, confirming that the pipeline is robust to precise weight choices.

I. Community Detection Comparison

To validate our choice of Louvain + clique refinement, we compare against three alternative community detection methods on the same graph (τ=6.0). Table VIII shows that our method produces 3× more communities (9 vs. 3), with 48% higher coherence (4.94 vs. 3.33) and nearly 2× the venue diversity (3.8 vs. 2.0) compared to pure Louvain.

Pure Louvain produces only 3 small communities because its default resolution groups most nodes into a single partition; our clique refinement breaks these into coherent subgroups. K-core decomposition [26] fails entirely at τ=6.0. Only our

TABLE VI: Runtime breakdown. Stages 2–3 (graph-algorithmic) complete in <0.4s; Stage 4 requires ∼70s of LLM calls. Extraction dominates at 99.9%.

Stage	Complexity	Time	Note
1. Extraction	$O(n)$	18.2h	I/O-bound (cloning)
2. Graph build	$O(n \times k)$	0.19s	inv. index: 91% red.
3. Communities	$O(n \log n)$	0.17s	Louvain + refine
4. Hypotheses	$O(K)$	∼70s	10 LLM calls
Total		**18.2h**	99.9% in Stage 1

TABLE VII: Edge weight component ablation at τ=6.0. Each row removes one component and rebuilds the graph. Edge counts and ShB differ from Table II because this ablation uses simplified pairwise computation without metric-weight terms; relative differences between rows are the relevant comparison.

Configuration	Edges	Δ	#C	ShB	Mean
Full model	**23,861**	–	**10**	**5/10**	**3.47**
− Benchmarks	12,966	−46%	10	0/10	3.43
− Methods	14,627	−39%	7	5/7	3.76
− Parameters	15,113	−37%	8	4/8	3.41
− Ablations	14,996	−37%	10	6/10	4.04

method achieves sufficient coverage (9 communities, 3.8 venues) for comprehensive hypothesis generation.

J. Hypothesis Analysis

Novelty verification. We sampled 20 hypotheses and searched Google Scholar and Semantic Scholar, reviewing the top 20 results per query. A hypothesis was classified as *novel* if no published paper reports the proposed experiment, *partial* if related work exists with different combinations, or *known* if already published. Of 20, 16 (80%) were novel, 4 (20%) partially overlapped, and none were already published. The 4 partial overlaps involved cases where the *technique* had been applied but on a different benchmark, or the *benchmark* had been studied but with a different method, precisely the gaps the system is designed to identify.

Gap types. Untested transfers account for 42%, unexplored combinations 19%, missing ablations 13%, and parameter optimization 12%. Cross-paper synthesis types (transfers + combinations) account for 61%, confirming that graph-structured communities enable genuine multi-paper reasoning rather than single-paper ablation suggestions.

Confidence calibration. Models self-rate 16% as high confidence, 35% medium, and 49% low, a conservative distribution where high-confidence hypotheses are backed by strong multi-paper evidence.

Community-level diversity. Within-community Jaccard similarity (0.154) is 2.04× higher than between-community (0.075). Under 100 random community assignments, the ratio averages 1.00±0.06 (z=16.8, p <0.001), confirming genuine topical coherence rather than a structural artefact.

K. Experimental Validation of Generated Hypotheses

To move beyond expert ratings, we ran 8 system-generated hypotheses as actual experiments across four domains (CV, graph mining, NLP, PEFT). Table IX summarises the results.

Four of eight hypotheses validated, three partially validated with overestimated effect sizes, and one failed due to architectural incompatibility. Methods tested include GAT [27] + DropEdge [28], LoRA [29] LR transfer, and RealNet [22] resolution (failed).

Validated hypotheses. LR tuning correctly identified learning rate as high-leverage: fine-tuning BLIP on COCO at LR=1e-05 achieved the best val loss (0.887), while LR=1e-04 caused the model to hallucinate entirely wrong captions (e.g., "a cat on a couch" for a fruit image). The system identified the right parameter but missed catastrophic failure modes. The LoRA experiment revealed a missed interaction: optimal LoRA LR (3e-04) is 15× higher than for full fine-tuning (2e-05), because fewer parameters tolerate more aggressive updates. LoRA at its best LR (PPL=13.67) actually outperformed full fine-tuning (PPL=13.92) with 0.65% of the parameters. Temperature scaling [30] reduced calibration error by 74% (ECE: 0.179→0.047) with zero accuracy loss. The selective data hypothesis confirmed that quality-filtered training texts outperform random sampling (PPL 13.36 vs. 13.47), though naïvely selecting the longest texts degrades performance catastrophically (PPL 204.12) because the longest entries are tables and code snippets.

Partial and failed hypotheses. The resolution ablation predicted 15–20pp drop from halving CLIP's input resolution, but observed only 2.6pp; modern ViTs are far more resolution-robust than predicted. The GAT+DropEdge hypothesis showed DropEdge helps 4-layer GAT on CiteSeer (+0.8%) but not Cora [31], and 8-layer models collapsed to ∼29% accuracy, indicating oversmoothing in attention-based GNNs differs from the depth problem in GCNs. The RealNet resolution hypothesis failed because the UNet has skip connections hardcoded for

TABLE VIII: Community detection comparison (τ=6.0). Clique refinement yields more, larger, more diverse communities. Uses the 9-community run of Section V-C (C10 excluded due to size).

Method	#C	Coh	$\bar{n}$	ShB	$\bar{V}$
Louvain + clique (ours)	**9**	**4.94**	**27**	**9/9**	**3.8**
Pure Louvain	3	3.33	5	3/3	2.0
Label propagation [25]	2	3.50	7	2/2	2.0
K-core decomposition [26]	0	–	–	–	–

TABLE IX: Experimental validation of system-generated hypotheses. "Validated" = predicted effect observed; "Partial" = correct direction but wrong magnitude; "Failed" = experiment could not run or prediction wrong.

Hypothesis	Dom.	Result	Key Finding
LR tuning for VLM	CV	Valid.	LR=1e-05 best
Resolution ablation	CV	Partial	2.6pp not 15–20pp
RealNet resolution	CV	Failed	UNet hardcoded
GAT + DropEdge	Graph	Partial	+0.8% CiteSeer
Decoding × context	NLP	Partial	Context helps
LoRA LR transfer	PEFT	Valid.	15× higher LR
Selective data quality	NLP	Valid.	Quality > random
Temp. calibration	CV	Valid.	74% ECE↓

TABLE X: Prompt template ablation. Evidence structure improves specificity (+1.40) while groundedness remains stable across templates.

Template	Mean	Spec	Grnd	Actn
Structured evidence	**3.85**	**4.40**	2.80	**3.80**
Chain-of-thought	3.60	3.40	**3.00**	3.60
Minimal (no evidence)	3.45	3.00	2.80	3.40
Negative framing	3.45	3.40	2.80	3.40

256×256; changing resolution causes tensor size mismatches at concatenation layers, a constraint undetectable from config files.

Patterns across domains. Three patterns emerge: (1) hypotheses transferring *regularization principles* (calibration, data quality, LR range) validate reliably across domains; (2) effect size predictions are systematically overestimated, consistent with our calibration analysis (31% predict >15%); (3) *architecture-specific constraints* (hardcoded dimensions, parameter-count-dependent optima) are invisible to config-level extraction. Across all 8 experiments, 87.5% of predictions are directionally correct but only 37.5% predict the correct magnitude.

Prompt template ablation. To isolate the prompt's contribution, we tested four templates on the same community with the same judge (Table X). Structured evidence achieves the highest mean (3.85) and specificity (4.40 vs. 3.00 for minimal), a +1.40 gap from ★-marked numbers. Groundedness is nearly identical across templates (2.80–3.00), confirming this dimension reflects citation confidence rather than actual evidence quality. Negative framing performs identically to minimal prompting (both 3.45), indicating the presence of evidence matters more than prompt framing.

VI. Discussion

Why community-scoped generation works. Scoping LLM reasoning to coherence-filtered communities, where all members share a benchmark or method, ensures every paper in context is experimentally relevant. The threshold ablation confirms this: at τ=6.0, all 10 communities share a concrete benchmark, producing peak groundedness (3.40) and actionability (3.86).

Domain applicability. We validated on 70 DM repositories from KDD (47), ICDM (12), CIKM (7), and WSDM (4), covering graph neural networks (GraphSAGE, GAT, GraphGPS), time series forecasting (iTransformer, PatchTST), tabular learning (TabPFN, FT-Transformer), and recommendation (LightGCN, RecBole). Extraction yielded 29 repos with 423 insights (41% yield), matching the main corpus (44.9%). When merged, 27 DM papers join cross-venue communities spanning 5 domains and up to 9 venues, e.g., KDD time series papers connect with NeurIPS/ICLR papers studying the same forecasting benchmarks. DM-specific hypothesis generation produces hypotheses spanning graph NNs, time series, tabular learning, and recommendation, confirming pipeline applicability beyond CV.

Complementarity with per-paper methods. B3 and our method serve fundamentally different purposes: B3 asks "what is missing from *this* paper?" while our method asks "what would happen if we combine findings from Papers A and B?" This question is impossible for any single-paper method. These are complementary, as recent benchmarks for scientific ideation [12] also recognise.

Baseline confound and judge sensitivity. To isolate contributions, we gave the standalone LLM the *same* community papers (B1-community) and random batches (B1-random), comparing against our structured prompting. The +0.66 advantage in Table I reflects the full pipeline vs. B1 with random inputs (as deployed in practice); the controlled comparison below decomposes *why*. Under the standard judge (Model A), B1-random (4.11) slightly outscores B1-community (4.05) and ours (3.94). However, under a stronger judge (Opus 4.5), the ordering reverses: B1-community (3.17) > ours (3.05) > B1-random (2.95), with community detection contributing +0.22 over random and total improvement of +0.10. Structured prompting does not improve mean quality over B1-community under this judge (-0.12), but shifts the quality profile toward specificity (3.84 vs. 3.04). This judge sensitivity analysis reveals that LLM-as-judge rankings can invert depending on the judge's ability to detect hallucinated citations, a finding consistent with recent surveys on LLM-as-judge reliability [32] and with implications beyond this paper.

Error analysis. Of 12,303 repositories, 5,742 (46.7%) lack parseable configs (Hydra [24], dataclasses), 12.9% have too few for ablation patterns, and 8.5% fail to clone. An LLM-assisted audit estimates 62.5% benchmark false-negative rate, though only 14% of repos are affected.

Coverage bias. CV venues achieve 32–36% inclusion vs. NLP at 5–6%, reflecting configuration style differences. Extraction rates show no temporal decline (2024: 40.6% vs. 2023: 34.8%, ρ=0.36, p=0.39). At τ=6.0, 80.2% of papers are graph isolates; the bottleneck is graph sparsity, addressable by expanding the corpus. Swanson-style disjoint-literature discovery [14], which identifies papers that *should* share features but do not, is a complementary task.

Insight types. Of 40,681 insights, `parameter_range` dominates (56.4%); the most valuable `config_result_link` (3.5%) pairs configurations with measured outcomes.

Limitations. (1) *Extraction coverage*: 44.9% yield; Hydra/dataclass support could reach 65–70%. (2) *Domain bias*: CV dominates; 27-keyword vocabulary misses ∼25 benchmarks, affecting 14% of repos. (3) *Evaluation gap*: LLM-judged dimensions assess structural quality, not plausibility (3.0/5 per human raters). A calibration check on 36 quantitative predictions shows 56% predict 0.1–5% (plausible), 14% predict 5–15% (optimistic), and 31% predict >15% (likely implausible). (4) *Threshold*: τ=6.0 is optimal here; may vary by domain. (5) *Scalability*: Stage 1 accounts for 99.9% of runtime. (6) *Venue coverage*: 13 of 28 venues appear at τ=6.0. (7) *Prompt–judge circularity*: ★-marked evidence in prompts likely inflates groundedness scores. (8) *Hypothesis granularity*: The system frequently proposes hyperparameter transfers (e.g., learning rate, batch size) rather than module-

level architectural combinations. While hyperparameter transfers are experimentally valid, they represent a less impactful form of scientific insight than novel architectural synthesis.

VII. Conclusion and Future Work

We presented a pipeline for cross-paper hypothesis generation grounded in experiment-similarity graphs mined from code. Evaluation shows gains over non-graph baselines (+0.66–1.00); a controlled decomposition identifies community scoping as the primary contributor, with structured prompting adding specificity (+1.40) but not overall quality. Validation across four domains confirms 87.5% directional accuracy but 37.5% magnitude accuracy: the system reliably identifies *what* to experiment with but overestimates *how much* it will help. Six experts rate the system useful with filtering (4/6 would adopt).

Key insight. Code repositories contain structured experimental knowledge invisible to text-based methods. By encoding this knowledge as graph edges, we enable hypotheses that synthesize across venues and domains, but the system remains an idea-generation aid, not an autonomous hypothesis engine.

Future work. (1) Expanding extraction to Hydra/dataclass configurations (estimated 65–70% coverage). (2) Parsing training logs for metric evidence. (3) Architecture-aware feasibility checking via static code analysis. (4) Temporal graph analysis for trend-aware generation.

Reproducibility

Code and data: https://github.com/dnbresearch/VIVEKA. Single A100 GPU; LLM calls use temperature=0.0 (variation across the 5 runs in Section V-C arises from non-deterministic API sampling at the provider side); experimental validation uses 3–10 random seeds per configuration on public datasets.

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